

PHARMACOVIGILANCE

Virtual Task Summary: Causality Assessment of Adverse Drug Reactions (ADRs)

Video reference: YouTube video titled "Causality Assessment" (provided by the student).

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1. Introduction

Pharmacovigilance is the science and activities concerned with detecting, assessing, understanding and preventing adverse effects or other medicine-related problems. Causality assessment is an important part of pharmacovigilance because an adverse event occurring after a medicine is taken does not automatically mean that the medicine caused the event.

2. What is Causality Assessment?

Causality assessment is a systematic process used to estimate the likelihood that a suspected medicine caused or contributed to an adverse drug reaction. It helps separate a possible drug-related reaction from an event that may be explained by the underlying disease, another medicine, a medical condition, or other factors.

3. Objectives

- To evaluate the relationship between drug exposure and an adverse event.
- To identify possible drug-related safety signals.
- To support clinical decision-making and patient safety.
- To improve the quality and usefulness of ADR reports.
- To support regulatory and risk-management decisions.

4. Important Factors Considered

The assessment generally considers the timing of drug administration and the event, the known pharmacological effects of the drug, improvement after withdrawal (dechallenge), recurrence after re-exposure (rechallenge), alternative causes, concomitant medicines, laboratory or clinical evidence, and previous knowledge about the reaction.

5. Common Causality Assessment Methods

Method	Main feature
WHO-UMC system	Uses clinical information and expert judgement to classify the relationship between medicine and event
Naranjo Algorithm	A structured questionnaire/scoring approach using factors such as timing, dechallenge, rechallenge and
Karch and Lasagna	A structured approach considering timing, alternative explanations, withdrawal response and other ev
French method	Uses chronological and semiological criteria to assess the relationship.

6. WHO-UMC Causality Categories

Certain: A clear time relationship exists; other explanations are unlikely, and the reaction is clinically or pharmacologically plausible. Rechallenge, when applicable, may provide strong supporting evidence.

Probable/Likely: The timing is reasonable, other causes are less likely, and the response to withdrawal is clinically reasonable; rechallenge is not usually required.

Possible: The timing is reasonable, but other diseases or medicines could also explain the event.

Unlikely: The timing makes a drug relationship improbable and another explanation is more likely.

Conditional/Unclassified: More information is required to complete the assessment.

Unassessable/Unclassifiable: The available information is insufficient or contradictory and cannot reasonably be supplemented.

7. Dechallenge and Rechallenge

Dechallenge means observing what happens after the suspected medicine is stopped. Improvement can support a causal relationship, although it is not proof by itself.

Rechallenge means the adverse event recurs after the medicine is given again. A positive rechallenge can provide strong evidence, but intentional rechallenge is not normally performed when it could put a patient at risk.

8. Example

Suppose a patient develops a rash soon after starting a new medicine. The healthcare professional reviews the timing, checks for other medicines or illnesses that could cause the rash, considers whether the reaction is known for the medicine, and observes the result after withdrawal when clinically appropriate. Based on the complete evidence, the case may be categorized as certain, probable/likely, possible, unlikely, or another appropriate category.

9. Importance in Pharmacovigilance

Causality assessment helps convert individual ADR reports into useful safety information. When many reports are collected and evaluated, patterns may reveal safety signals that can lead to further investigation, updated warnings, changes in monitoring, or other risk-minimisation actions. Therefore, accurate reporting and good-quality clinical information are essential.

10. Pharmacovigilance in India

In India, the Pharmacovigilance Programme of India (PvPI), coordinated through the Indian Pharmacopoeia Commission, supports monitoring of medicine-related adverse events. Healthcare professionals and patients can report suspected ADRs through the PvPI system and ADR Monitoring Centres. Suspected ADRs are reported even when causality has not yet been proven.

11. Conclusion

Causality assessment is a key component of pharmacovigilance. It does not simply ask whether an event happened after a medicine was used; it systematically examines timing, clinical evidence, alternative explanations and response to withdrawal or re-exposure. Proper causality assessment supports safer medicine use, better patient care and effective drug-safety monitoring.

References

1. World Health Organization (WHO) – Pharmacovigilance and Programme for International Drug Monitoring.
2. Indian Pharmacopoeia Commission (IPC) – Pharmacovigilance Programme of India (PvPI), FAQ and ADR reporting resources.
3. CDSCO – Post Marketing Drug Safety Monitoring and pharmacovigilance guidance documents.
4. Provided YouTube learning resource: "Causality Assessment".

Note: This is an assignment-ready educational summary based on the identified video topic and standard pharmacovigilance references. It is intended for academic use.